

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
April - May 2016

Max. Marks: 100

Class: S.E

Name of the Course: Discrete Structures

Course Code: CSC305

Duration: 3 hours

Semester: III

Branch: Computer Engineering

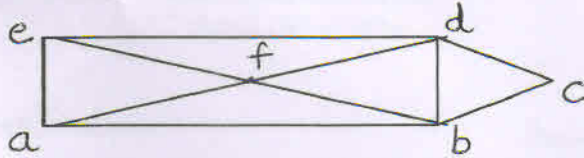
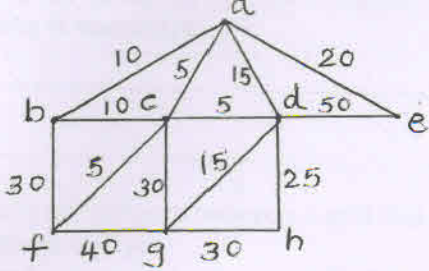
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Instructions:

- (1) All Questions are Compulsory
- (2) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Find the number of integers between 1 and 500 which are divisible by i) Two, three or five ii) Neither by two nor by three nor by five.	10M
Q 1 (a)	OR Explain the following terms w.r.t set theory a) Symmetric Difference b) Union of Sets c) Universal Sets d) De Morgans Laws	10M
Q 1 (b)	Let $A = \{a, b, c, d, e, f, g, h\}$. Consider the following subsets $A_1 = \{a, b, c, d\}$ $A_2 = \{a, c, e, g, h\}$ $A_3 = \{a, c, e, g\}$ $A_4 = \{b, d\}$ $A_5 = \{f, h\}$ Determine whether each of the following is partition of A or not. Justify your answer (i) $\{A_1, A_2\}$ (ii) $\{A_1, A_5\}$ (iii) $\{A_3, A_4, A_5\}$ (iv) $\{A_2, A_5\}$ (v) $\{A_3, A_2, A_4\}$ OR	10M
Q 1 (b)	Out of 250 candidates who failed in either in Mathematics or Physics or in Aggregate, it was revealed that 128 failed in Mathematics, 87 in Physics and 134 in Aggregate, 31 failed in Mathematics and in Physics, 54 failed in Aggregate and Mathematics, 30 failed in Aggregate and physics. Find: How many failures (i) In all the three subjects (ii) In Mathematics but not in Physics (iii) In Aggregate but not in Mathematics (iv) In Physics but not in Aggregate or Mathematics	10M

Q2 (a)	Construct truth tables to determine whether each of the following is a tautology, contingency, or a contradiction (i) $(p \wedge q) \rightarrow p$ (ii) $(p \rightarrow q) \leftrightarrow (q \vee \neg p)$ (iii) $(p \wedge (\neg p \vee q)) \wedge \neg q$	10M
Q2 (b)	Prove by mathematical induction that $9^{2n} - 1$ is divisible by 80 for all n belongs to N.	10M
Q3 (a)	Let $a = \{1, 2, 3, 4, 5\}$ and R be the relation defined by aRb if and only if $a < b$, compute R, R^2, R^3 . Draw digraph of R, R^2, R^3 OR	10M
Q3 (a)	Let $A = \{1, 2, 3, 4\}$. $R = \{(1, 2), (1, 1), (1, 3), (2, 4), (3, 2)\}$ and $S = \{(1, 4), (1, 3), (2, 3), (3, 1), (4, 1)\}$ Find $SoR, RoS, M_{RoS}, M_{SoR}$	10M
Q3 (b)	Explain equivalence relation. Let $A = \{1, 2, 3, 4, \dots, 14, 15\}$. Consider the equivalence relation R defined on $A \times A$ by $(a, b) R (c, d)$ if $ad = bc$. Find the equivalence class of $(3, 2)$. OR	10M
Q3 (b)	If $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 4), (2, 1), (2, 2), (3, 3), (4, 4)\}$. Find the transitive closure of R using Warshall's Algorithm.	10M
Q4 (a)	Define the types of functions with suitable examples (i) Surjective Function (ii) Injective Function (iii) Bijective Function (iv) Everywhere defined function OR	10M
Q4 (a)	What do you understand by Pigeon Hole Principle? Also explain Extended Pigeon Hole Principle. Solve for if 30 dictionaries in a library contain a total of 61327 pages, then show that one of the dictionary has at least 2045 pages.	10M
Q4 (b)	Let $f: R \rightarrow R, f(x) = x^2 - 1$ and $g(x) = 4x^2 + 2$. Find (i) $g \circ f$ (ii) $f \circ g$ (iii) $f \circ f$ (iv) $f \circ (g \circ f)$ (v) $g \circ (f \circ g)$ OR	10M
Q4 (b)	Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions. (i) If f and g are injective then $g \circ f$ is injective. (ii) If f and g are surjective then $g \circ f$ is surjective. (iii) If f and g are bijective then $g \circ f$ is bijective.	10M

Q5 (a)	Define Euler Path, Euler Circuit, Hamiltonian Path and Hamiltonian Circuit. Also find Hamiltonian path and Hamiltonian circuit for the below graph given. 	10M
Q5 (b)	State Prim's Algorithm and find minimum spanning tree for the following graph. 	10M